

Lesson 22 Inverse Function and Function Composition

1. For each pair of functions f and g , find $f \circ g$ and $g \circ f$ whenever they are defined. If a composition is not defined, explained why.
 - a) $f: \mathbb{Z} \rightarrow \mathbb{Z}, f(x) = 3x - 2$, and $g: \mathbb{Z} \rightarrow \mathbb{Z}, g(x) = |x|$
 - b) $f: \mathbb{N} \rightarrow \mathbb{R}, f(x) = \ln x$, and $g: \mathbb{R} \rightarrow \mathbb{Z}, g(x) = \lfloor x \rfloor$
 - c) $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^2 + x$, and $g: \mathbb{R} \rightarrow \mathbb{R}, g(x) = e^{-x^2}$
 - d) $f: \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}, f(x, y) = xy$, and $g: \mathbb{Z} \rightarrow \mathbb{Z} \times \mathbb{Z}, g(x) = (x, |x|)$
 - e) $f = \{(1,2), (3,4), (5,6), (7,8)\}$ and $g = \{(2,7), (4,5), (6,3), (8,1)\}$
 - f) $f = \{(1,2), (3,4), (5,6), (7,8)\}$, and $g: \mathbb{Z} \rightarrow \mathbb{Z}, g(x) = 2x$.
2. Let $f: \mathbb{Z} \rightarrow \mathbb{Z}$ and $g: \mathbb{Z} \rightarrow \mathbb{Z}$.
 - a) If $f(x) = 5 - x$, and $(g \circ f)(x) = x + 5$, what is $g(x)$?
 - b) If $g(x) = 2x + 4$, and $(g \circ f)(x) = 6x$, what is $f(x)$?
3. Let A, B, C be sets, and suppose $f: A \rightarrow B$ and $g: B \rightarrow C$.
 - a) If $g \circ f$ is one-to-one, then f is one-to-one.
 - b) If $g \circ f$ is onto, then g is onto.
4. Suppose $f: A \rightarrow B$ is a bijection. Prove that:
 - a) $f \circ i_A = f$ (similarly, $i_B \circ f = f$)
 - b) $f^{-1} \circ f = i_A$ (similarly, $f \circ f^{-1} = i_B$)

The following exercises are optional.

5. Let $f: A \rightarrow B$ and $g: B \rightarrow A$ be bijections. Show that if $g \circ f = i_A$, then $f^{-1} = g$.
6. Let A, B, C be sets, and suppose $f: A \rightarrow B$ and $g: B \rightarrow C$. If f and g are bijections, then by #3a and #3b, $g \circ f: A \rightarrow C$ is a bijection and is therefore invertible. Prove that $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.